Factor-base designs: G1, the m = 4 series on p_1 (run-20261016T0558Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md Amendment 5 (registered as "Amendment 4" in commit `113f31659`, before any G1 row; renumbered when the lanes were merged, see the note at the end), with Amendments 1–3 for everything it does not change.

Run

- **Command:** `target/release/examples/fb_designs_slope run --arms A0,A11 --m 4 --t-min 4 --budget 3600 --grace 300 --vmem-kb 15000000 --distinct-targets --file 'G1-{arm}-m{m}' --out-dir research/factor_base_designs_20261015/runs`
- **Commit measured:** `113f31659af224c2e8eb46ebf86e5a18fb0993a3`, recorded in every row.
- **Host:** "vm", Intel Xeon @ 2.10 GHz, 4 cores, 15 GB RAM, no swap, a Claude Code cloud container. **Isolation:** one system at a time.
- **Window:** 2026-10-16 05:58Z to about 06:39Z (UTC).
- **Limits:** F4 budget 3600 s, hard kill 3900 s, `ulimit -v 15000000` per child. The session also runs inside a memory cgroup that the harness does not set. Its OOM killer ended two systems at about 14 GB resident (below).
- **Seeds:** as in Amendment 2, with G2 on (`--distinct-targets`). G2 redrew nothing (`target_redraws` = 0 on every row). `compare-rows` confirms that the t = 4 systems are the F4 systems that stopped at 300 s (`runs/G1-vs-F4-crosscheck.txt`).
- **Records:** `runs/G1-{A0,A11}-m4.jsonl` and `.log`, `runs/analysis-g1.txt` (native `analyze --g1`) and `runs/G1-oom-kernel-log.txt`.

Graph

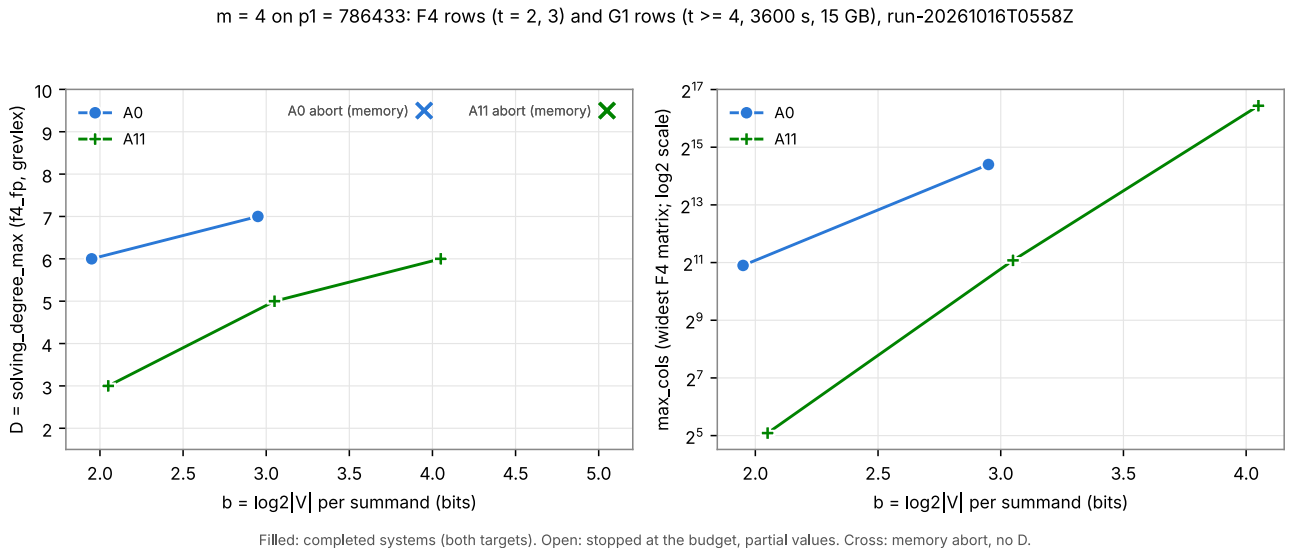


Figure 1: m = 4 on p_1: D (left) and widest F4 matrix (`max_cols`, $\log_2$ scale, right) against b, for A0 and A11. t = 2, 3 come from F4 (300 s budget) and t ≥ 4 from G1 (3600 s, 15 GB). Both targets are plotted (n = 2 per cell; they coincide). Crosses mark memory aborts, which have no D. Rows from `runs/F4-*-m4.jsonl` and `runs/G1-*-m4.jsonl`, plotted by `plot.py`.

Every system in the m = 4 series

arm	t	source	vars	eqs	D	max_cols	time (s)	status
A0	2	F4	10	11	6 / 6	1911 / 1911	0.3 / 0.3	ok, ok
A0	3	F4	14	15	7 / 7	21619 / 21619	96.4 / 93.8	ok, ok
A0	4	G1	18	19	– / –	– / –	404.0 / 406.6	aborted, aborted
A11	2	F4	21	22	3 / 3	34 / 34	0.0 / 0.0	ok, ok
A11	3	F4	28	29	5 / 5	2159 / 2159	0.3 / 0.3	ok, ok
A11	4	G1	35	36	6 / 6	88740 / 88740	751.9 / 738.6	ok, ok
A11	5	G1	42	43	– / –	– / –	73.5 / 65.9	aborted, aborted

- **A0, t = 4:** both systems aborted on an allocation failure under `ulimit -v 15000000` (SIGABRT), at about 405 s. F4 had stopped the same systems at 300 s, at partial D = 7.
- **A11, t = 5:** both systems were killed with SIGKILL after 66–74 s. The kernel log shows the memory-cgroup OOM killer (`CONSTRAINT_MEMCG`), at 13.97 GB resident, below the 15 GB address-space limit. The same OOM event also killed an idle shell of the session that was waiting on the run. That shell was not part of the measurement.

Verdicts (Amendment 5, G1; §1 thresholds)

arm	completed cells (t)	σ	$\Delta\log_2$ max_cols/Δb	verdict
A0	2 (2, 3)	1.0000	3.500	inconclusive (fewer than 3 cells; t = 4 aborted for memory)
A11	3 (2–4)	1.5000	5.675	no middle ground at this arity (σ ≥ 0.5)

- **Prediction "A0′s σ ≥ 0.5 at m = 4":** not decided (2 completed cells). Amendment 5 makes no further budget or memory change without a new amendment.
- **A11:** no prediction was registered. Its m = 4 series is D = 3, 5, 6 over t = 2–4.
- **Width (secondary, descriptive):** A11′s `max_cols` over A0′s is 0.018 at t = 2 and 0.100 at t = 3. There is no shared completed cell at t = 4.

Observations

- At m = 4, A11 starts below A0 on both D and width: D 3 vs 6 at t = 2, and 5 vs 7 at t = 3. But its D rises by 3 over two bits, and its width grows about 5.7 bits per bit of factor base, against A0′s 3.5 over its two cells.
- A11′s partial D at F4′s 300 s stop (6) already equalled its final D at t = 4. The 3600 s budget changed its status, not its degree.
- The binding limit at m = 4 is memory for both arms: A0 at t = 4, A11 at t = 5. On this host the effective ceiling is the session's memory cgroup (about 14 GB resident), not the `ulimit` the amendment raised.

Deviations

None in the protocol. The amendment's memory limit took effect as written, but a memory cgroup this lane did not set ended two systems below it. Amendment 5 already counts an OOM kill as an abort, and those rows record `child exited: signal: 9` as their reason. The kernel log is committed beside them.

Numbering and a parallel run (note added 2026-10-19)

Two amendments were registered independently as "Amendment 4" on 2026-10-16: this one (G1, G2) at 05:58Z, and C1–C4 on the experiments-2 lane at 06:16Z (commit `025746fd5`). The study's main line carries C1–C4 as Amendment 4, so this one is Amendment 5 in the merged preregistration. Its text is unchanged under a numbering note. The analysis file is `runs/analysis-g1.txt`, renamed from this lane's `analysis-amendment4.txt`, a name C1–C4 also uses; apart from the amendment number its content is the same.

C2 of that amendment ran the same m = 4, t = 4 systems as G1, with a 13 GB limit instead of 15 GB. `compare-rows` finds all four shared systems identical (`runs/G1-vs-C2-crosscheck.txt`): both A0 systems abort for memory under either limit, and both A11 systems complete with the same D and `max_cols`. The rows only G1 holds are the A11 t = 5 OOM kills.